

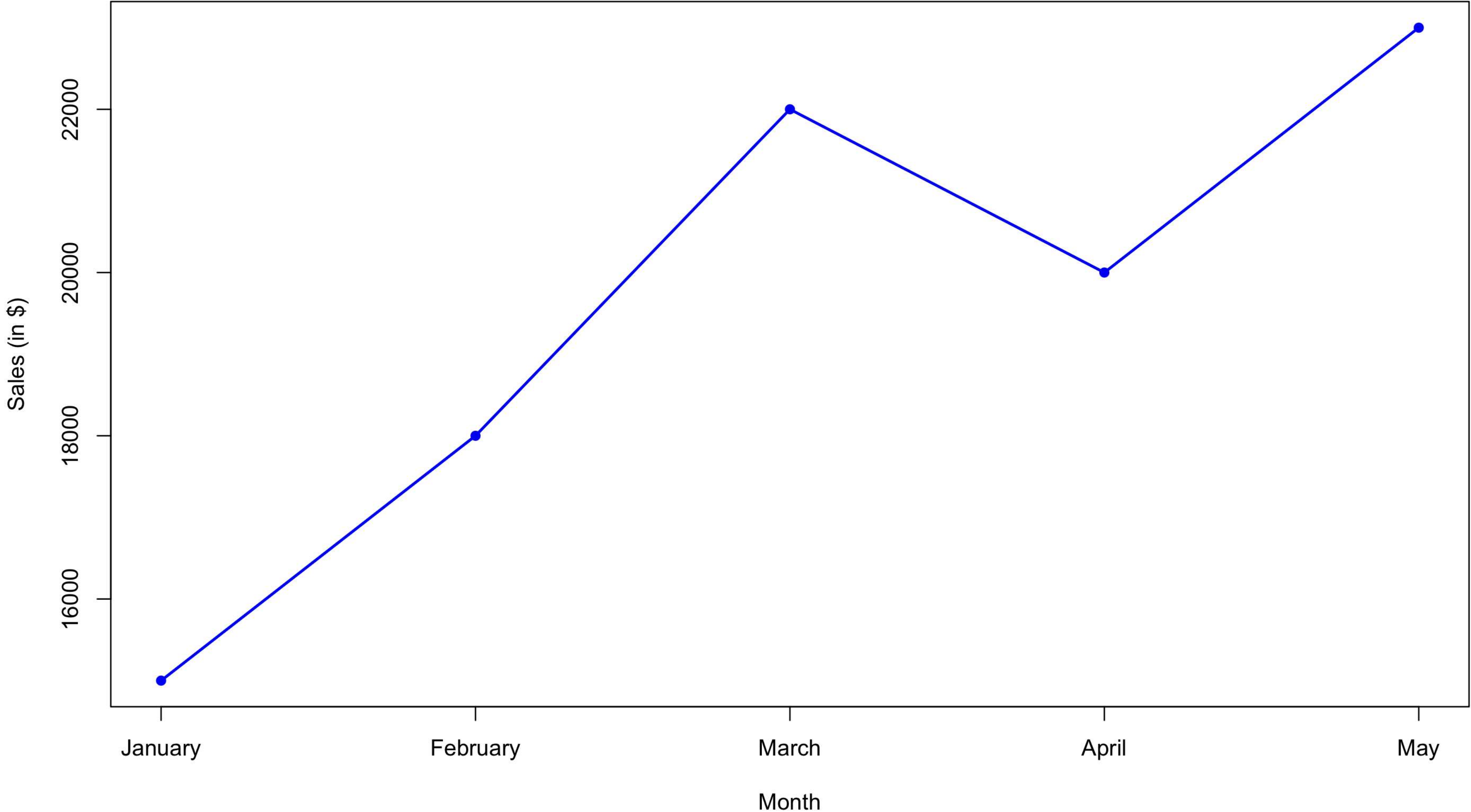


```
> # -----
> # Dataset Setup
> # -----
> sales_data <- data.frame(
+   Month = c("January", "February", "March", "April", "May"),
+   Sales = c(15000, 18000, 22000, 20000, 23000)
+ )
>
> # Sample advertising budget data (since not provided in dataset)
> sales_data$Ad_Budget <- c(2000, 2500, 3200, 2800, 3500)
>
> print(sales_data)
  Month Sales Ad_Budget
1 January 15000     2000
2 February 18000     2500
3   March 22000     3200
4  April 20000     2800
5    May 23000     3500
>
> library(ggplot2)
>
> new_window <- function() {
+   if (.Platform$OS.type == "windows") {
+     windows()
+   } else {
+     quartz()
+   }
+ }
>
> # -----
> # Q1: Time Series Line Chart
> # -----
> sales_ts <- ts(sales_data$Sales, start = 1, frequency = 1)
>
> new_window()
> plot(sales_ts,
+       type = "o",
+       col = "blue",
+       lwd = 2,
+       pch = 16,
+       main = "Monthly Sales Trend",
+       xlab = "Month",
+       ylab = "Sales (in $)",
+       xaxt = "n")
> axis(1, at = 1:5, labels = sales_data$Month)
>
> # -----
> # Q2: Scatter Plot - Advertising Budget vs Sales
> # -----
> new_window()
> print(
+   ggplot(sales_data, aes(x = Ad_Budget, y = Sales)) +
+   geom_point(size = 4, color = "darkgreen") +
```

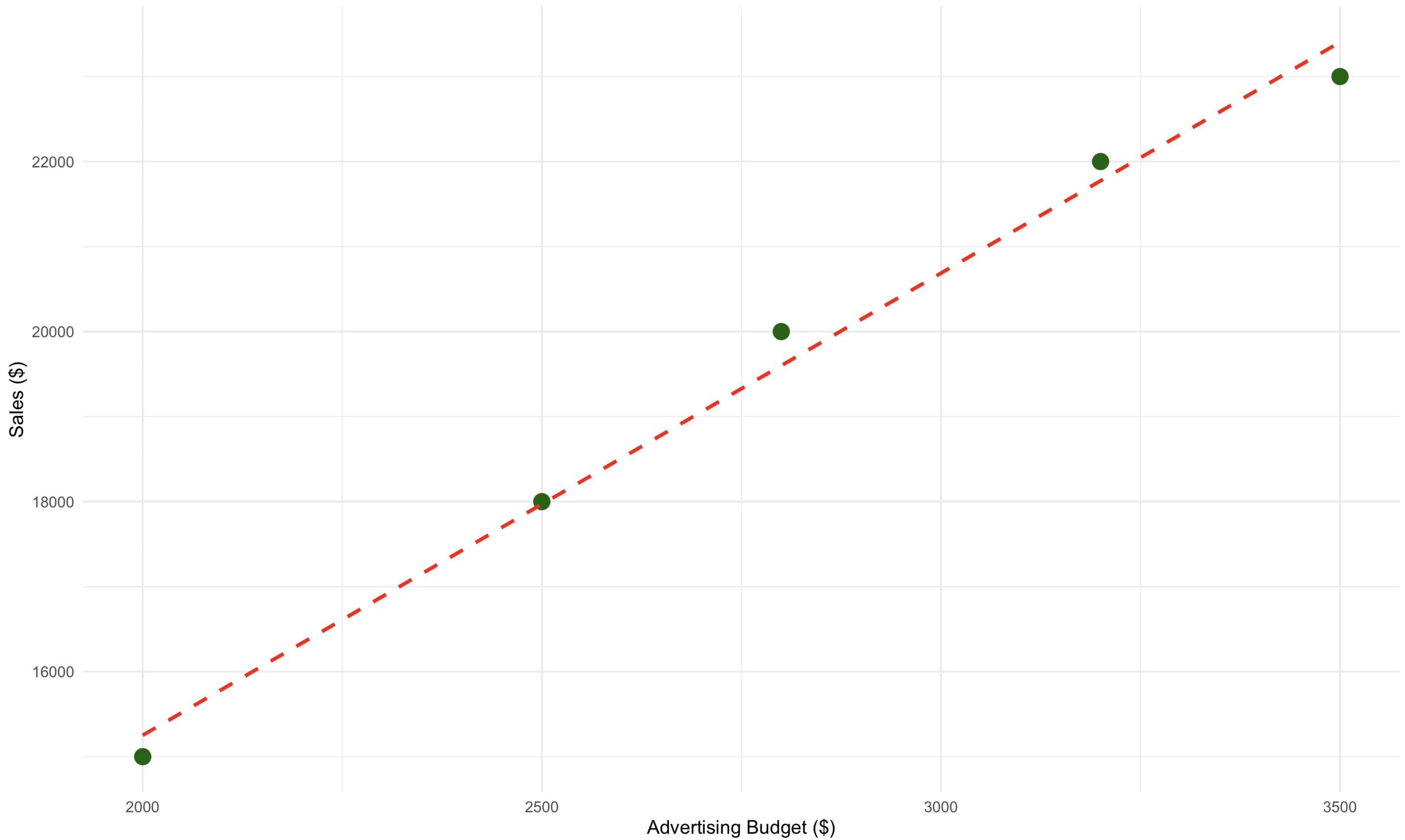


```
3   March 22000      3200
4   April 20000      2800
5     May 23000      3500
>
> library(ggplot2)
>
> new_window <- function() {
+   if (.Platform$OS.type == "windows") {
+     windows()
+   } else {
+     quartz()
+   }
+ }
>
> # -----
> # Q1: Time Series Line Chart
> # -----
> sales_ts <- ts(sales_data$Sales, start = 1, frequency = 1)
>
> new_window()
> plot(sales_ts,
+       type = "o",
+       col = "blue",
+       lwd = 2,
+       pch = 16,
+       main = "Monthly Sales Trend",
+       xlab = "Month",
+       ylab = "Sales (in $)",
+       xaxt = "n")
> axis(1, at = 1:5, labels = sales_data$Month)
>
> # -----
> # Q2: Scatter Plot - Advertising Budget vs Sales
> # -----
> new_window()
> print(
+   ggplot(sales_data, aes(x = Ad_Budget, y = Sales)) +
+     geom_point(size = 4, color = "darkgreen") +
+     geom_smooth(method = "lm", se = FALSE, color = "red", linetype = "dashed") +
+     labs(title = "Advertising Budget vs Monthly Sales",
+          x = "Advertising Budget ($)",
+          y = "Sales ($)") +
+     theme_minimal()
+ )
`geom_smooth()` using formula = 'y ~ x'
>
> # -----
> # Q4: Autocorrelation Plot (ACF)
> # -----
> new_window()
> acf(sales_data$Sales,
+     main = "Autocorrelation Plot of Monthly Sales",
+     lag.max = 4)
>
> |
```

Monthly Sales Trend



Advertising Budget vs Monthly Sales



Autocorrelation Plot of Monthly Sales

